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EVENT REPORT ON

"ROBERTS MECHANISM"

by

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1. INTRODUCTION TO ROBERTS MECHANISM

The **Roberts Four Bar Mechanism** is a fascinating topic in mechanical engineering! Here's a detailed overview:

Introduction

The Roberts Four Bar Mechanism, also known as the Roberts Linkage, is a type of four-bar linkage that converts rotational motion into approximate straight-line motion. It was developed by **Richard Roberts**, an English engineer, in the 19th century.

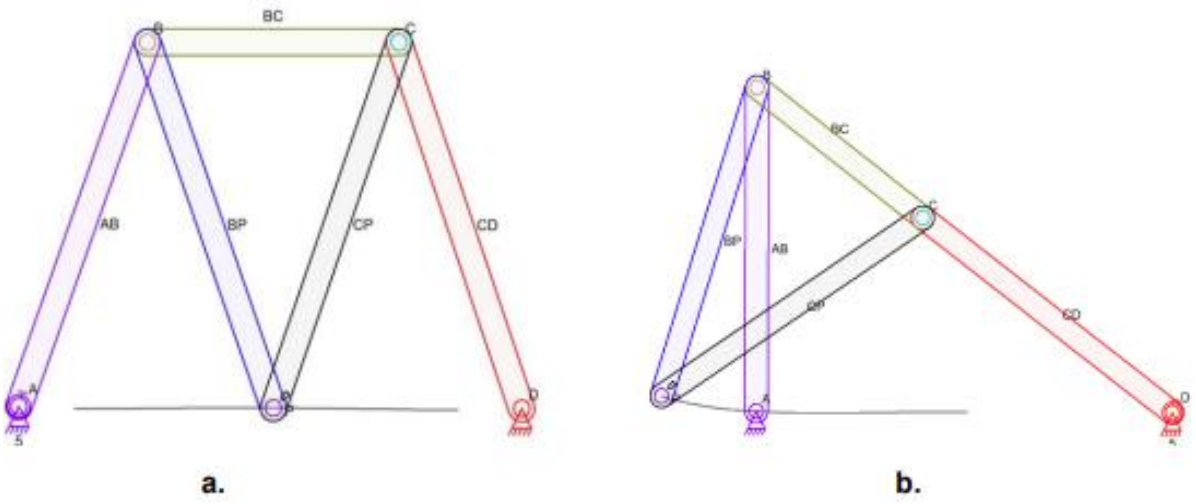


Figure 1 ROBERTS MECHANISM

Mechanism Overview

Robert's mechanism converts a rotational input motion into an approximate straight-line output motion. The rotating or oscillating input motion may come from one of the motors (A or B) connected to the cranks (AB and CD) which may swing to right until AB and BC are in a straight-line, and also swing back to the right until BC and DC are in a straight line. (Doughtie & James, 1954)..

Interesting Facts

- The Roberts linkage is a type of Watt-type linkage, named after James Watt, who also developed a similar mechanism for steam engines.
- It is classified as a Grashof rocker-rocker linkage, meaning it can achieve continuous rotation.
- The mechanism is symmetrical, which helps in achieving balanced and smooth motion.

Grashof's Theorem

This a fundamental principle in kinematics, especially relevant to four-bar linkages. Here's the theorem statement:

Grashof's Theorem states "**that for a planar four-bar linkage consisting of four links (one of which is fixed), the sum of the lengths of the shortest and longest links should be less than or equal to the sum of the lengths of the other two links in order for the linkage to have at least one link capable of a full 360-degree rotation relative to the fixed link**".

About Richard Roberts

- A further development was made by Richard Roberts (1789-1864). It is another example of a watt type linkage, and here the restrictions on link lengths can be relaxed.
- Roberts was born at L Llanymynech, Powys, on the border between England and Wales. He was the son of William Roberts, a shoemaker who also kept the New Bridge tollgate. Roberts was educated by the parish priest and early found employment with a boatman on the Ellesmere Canal and later at the local limestone quarries.
- He received some instruction in drawing from Robert Bough, a road surveyor, who was working under Thomas Telford. Roberts moved his business in 1821, to the Globe Works in Faulkner Street.
- Whilst there he improved a reed-making machine, originally invented by the American Jeptha Avery Wilkinson, and in 1822 he patented a power loom.
- This was made entirely of iron and being precision-made was able to operate at high speed. They were turned out at the rate of 4,000 per year by 1825. In 1824 he invented his most famous machine, the self-acting spinning mule, and patented it in March 1825.

Advantages:

1. **Simplicity**: The mechanism is relatively simple in construction and easy to manufacture.
2. **Symmetry**: It is a symmetrical four-bar linkage, which helps in achieving a balanced motion.
3. **Linear Guidance**: It provides a good approximation of straight-line motion, which is useful for guiding a point along a linear path.
4. **Versatility**: It can be used in various applications, such as guiding the piston in engines or in industrial robots.

Limitations:

1. **Approximate Motion**: The motion generated is an approximation of a straight line, not a perfect straight line.
2. **Complexity in Design**: While simple in construction, the design and calculation of the exact proportions of the linkage members can be complex.
3. **Limited Range**: The length of the straight-line segment is limited, and the mechanism may not be suitable for applications requiring long straight-line motion.
4. **Mechanical Wear**: Continuous use can lead to wear and tear, affecting the accuracy and efficiency of the mechanism over time.

2. FABRICATION OF ROBERTS MECHANISM

Design and Calculations

1. Conceptual Design:

Start by conceptualizing the design of the Roberts Four-Bar Mechanism. Determine the desired motion and dimensions of each link based on the required output motion.

2. Kinematic Analysis:

Perform kinematic analysis using equations to ensure the desired straight-line motion. The equations governing the four-bar mechanism are:

$$f(\theta_1, \theta_2, \theta_3, \theta_4) = 0$$

where:

1. θ_1 is the input angle of the crank,
2. θ_2 is the output angle of the rocker,
3. θ_3 and θ_4 are the angles of the connecting rods.

3. The Degrees of Freedom:

The kinematics of a four-bar linkage can be described using various mathematical equations, including the Kutz Bach criterion for calculating the degrees of freedom:

$$n = 3(l-1) - 2j - h$$

where n is the number of degrees of freedom, l is the number of links, j is the number of joints, and h is the number of higher pairs (sliding joints).

The mechanism used binary link which has 2 nodes and tertiary link which has 3 nodes. Binary links are AB and CD, and the tertiary link is the triangle BPC. It has 4 lower pairs which are A, B, C, and D and 0 higher pair. Degrees of freedom and mobility By using the Gruebler's Equation, we can compute the degrees of freedom of Robert's Mechanism

$$\mathbf{F = 3(n - 1) - 2l - h}$$

Robert's Mechanism consist of 4 links, 4 lower pairs and 0 higher pairs Solution:

n = 4, l = 4, h = 0 $\mathbf{F = 3(4 - 1) - 2(4) - 0 = 1}$

Therefore, the degrees of freedom of Robert's Mechanism is 1

4. Configuration

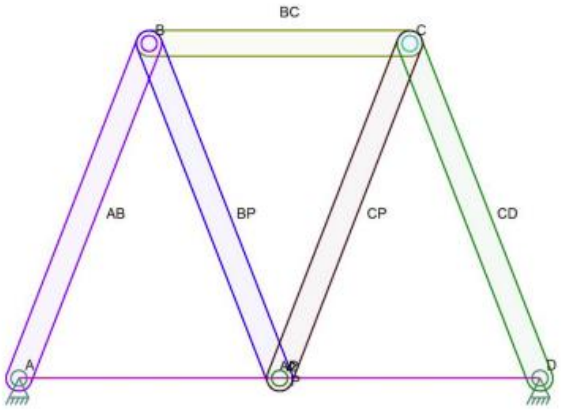


Figure 2: *Proper Arrangement of links in Roberts' Mechanism drawn using linkage application.*

Proper Arrangement of Links In figure 2 shows Robert's mechanism which is a four-bar linkage, the cranks AB and CD, and connecting link BP and CP are equal. The BPC blade is in the shape of an isosceles triangle which means two sides are equal: BP and CP. In addition, the crank AB is parallel and equal to the connecting link CP, also, crank CD and connecting link BP. Connecting rod BC is half as long as the distance between point A and point D

$$AB = BP = CP = CD \text{ and } AD = 2\ BC$$

Number of links, joints and pairs Based on figure 2, Robert's mechanism consists of 4 linkages, (1) ground or frame Point A and Point D, (2) Point A and Point B, (3) Point B and Point C and lastly (4) Point C and Point D.

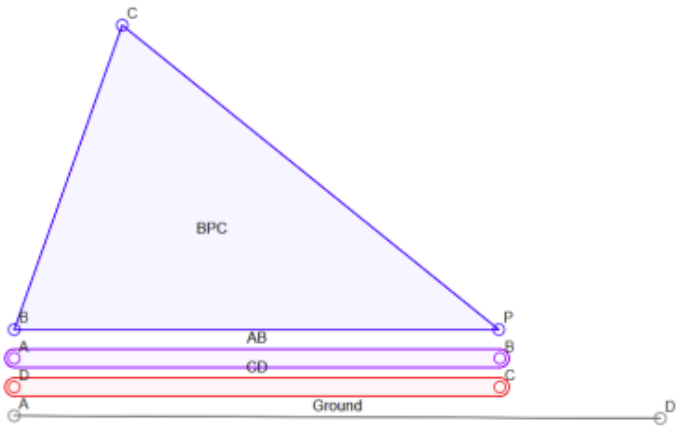


Figure 3: *Disassembly of the links of Robert's Mechanism.*

Components and Materials

1. **Links**: The links can be made of materials such as aluminium, steel, or plastic, depending on the load requirements and application.

2. **Joints**: Use revolute pairs for the joints, typically made of hardened steel or brass for smooth rotation.

3. **Fasteners**: Bolts, nuts, and washers to secure the components

Fabrication Process

1. **CAD Modelling**: Create a detailed 3D model of the mechanism using CAD software (e.g., SolidWorks, AutoCAD). This step is crucial for visualizing the design and ensuring all components fit together correctly.

2. **3D Printing**: If using 3D printing, export the CAD model to STL format and print the components using a suitable 3D printer. Use high-quality printing materials to ensure durability.

3. **Machining**: For traditional fabrication, use CNC machines or manual tools to cut and shape the links and joints. Precision is key to ensuring the mechanism operates smoothly.

4. **Assembly**: Assemble the components by fitting the links into the joints and securing them with fasteners. Make sure to follow the designed sequence to avoid misalignment.

Testing and Optimization

1. **Initial Testing**: After assembly, test the mechanism to check for smooth operation and any misalignments. Adjust the links and joints as needed.

2. **Load Testing**: Test the mechanism under load conditions to ensure it can withstand operational forces. Make **any necessary adjustments to improve performance.**

3. **Fine-Tuning: Optimize the de**sign and assembly for better efficiency and accuracy.

3. APPLICATION OF ROBERTS MECHANISM

Industrial Applications

1. Hand operated controllers

There are a lot of controllers used in various applications may it be for controlling operating remote systems such as trains or robots which follows a standard mechanism; however, this specific hand grip controller (Fig. 6) allows a linear motion using a modified Robert's mechanism. The modification is that by adding a second a linkage connected to the hand grip, the rotational movement produced by Robert's mechanism will be countered by the rotational movement produced on the hand grip by the added linkage.

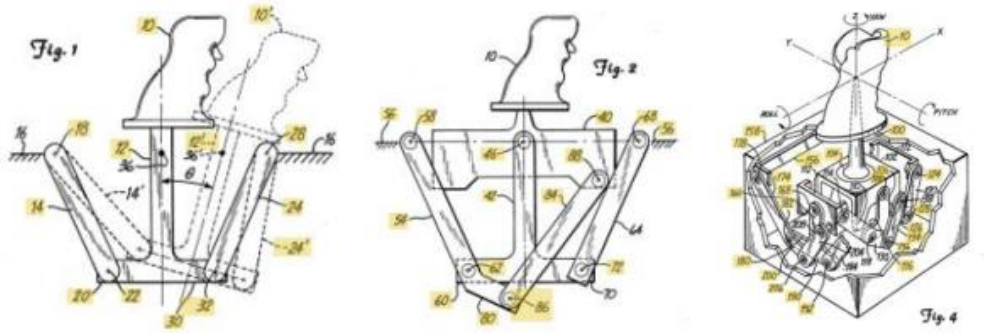


Figure 6: *Three degree of freedom translational axis hand controller mechanism by Israel Menahem and James Bacon*

2. Robotics

Robert's mechanism may be applied in the design and construction of robots. Shown below are the animation of a single project robot with a similar motion if not exactly, according to Robert's mechanism.

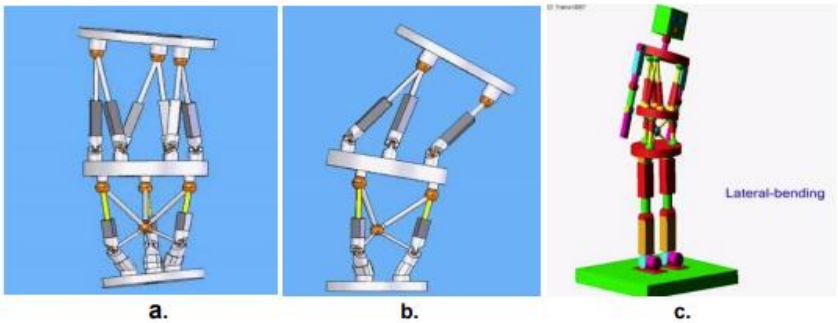


Figure 7: *3D Model of LARM Waist Trunk: (a) Walking, (b) Manipulation, (c) Lateral Bending Movement.*

3. Mechanical walkers

These are vehicles that walk on legs rather than wheels, thus the name walking vehicles. They are constructed with one to more than eight legs that provide trajectories using d

different properties. Various linkages are used as leg mechanisms, for instance, Trot Bot's and Jansens, and the example below shows a strider linkage that incorporates a four-bar walking linkage- which may also include Robert's mechanism.

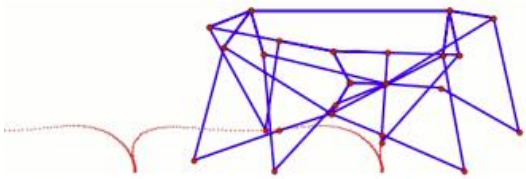


Figure 9: *Motion Path of Mechanical Walker.*

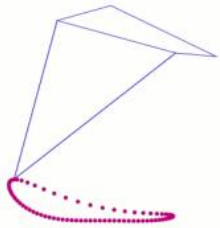




Figure 10: *Actual Bipedal Robot with application of mechanical walkers.*

Figure 8: *A Four-Bar Walking Linkage with a Shorter Rocker bar.*

4. Horizontal mill engine

Before the rise of electric motors in the 1930's, mill engines were a popular source of power for use industrially. They ran slowly, and usually the valves were operated through some form of "trip" gear allowing the steam to be cut off at a precisely defined point in the stroke. As a consequence, they were highly economical of steam and their speed could be closely governed. The horizontal mill engine below follows a straight-line motion using Robert's mechanism.



Figure 11: *Actual Working Horizontal Mill Engine*

5. RLRBML

A planar reconfigurable linear (rectilinear) rigid-body motion linkage (RLRBML) is a linkage that is able to produce linear rigid-body motion by transforming rotational motion. It is achieved by combining (a) Robert's linkage and a (b) double parallelogram linkage.

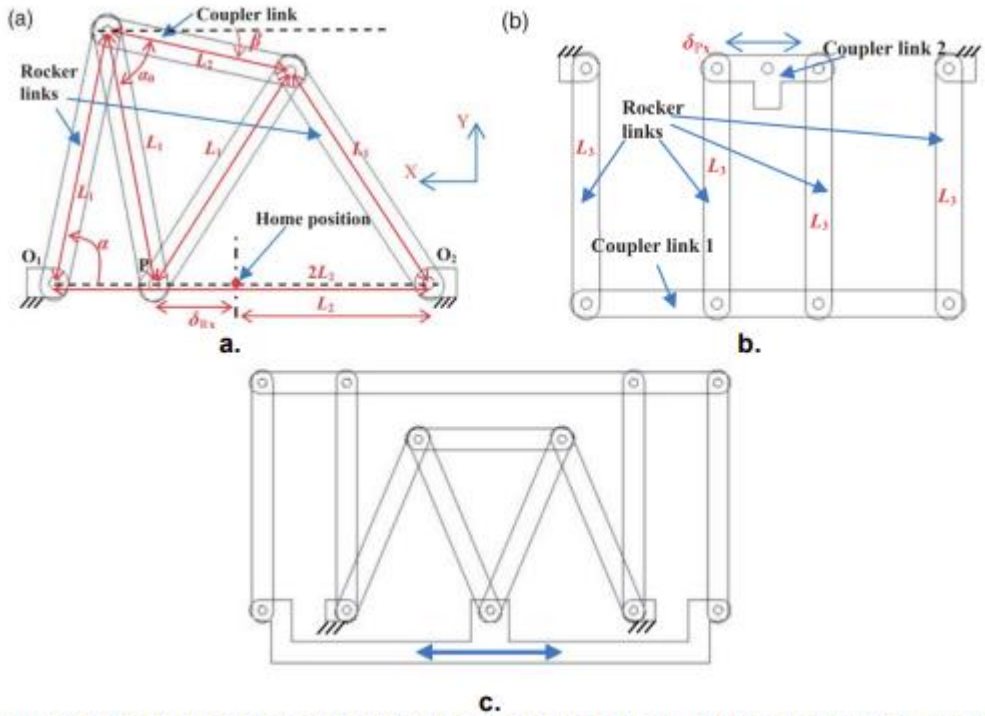


Figure 12: *Mechanism involved in RLRBML: (a) Robert's Linkage, (b) Double Parallelogram Linkage, (c) Combination of Robert's and Double Parallelogram Linkage.*

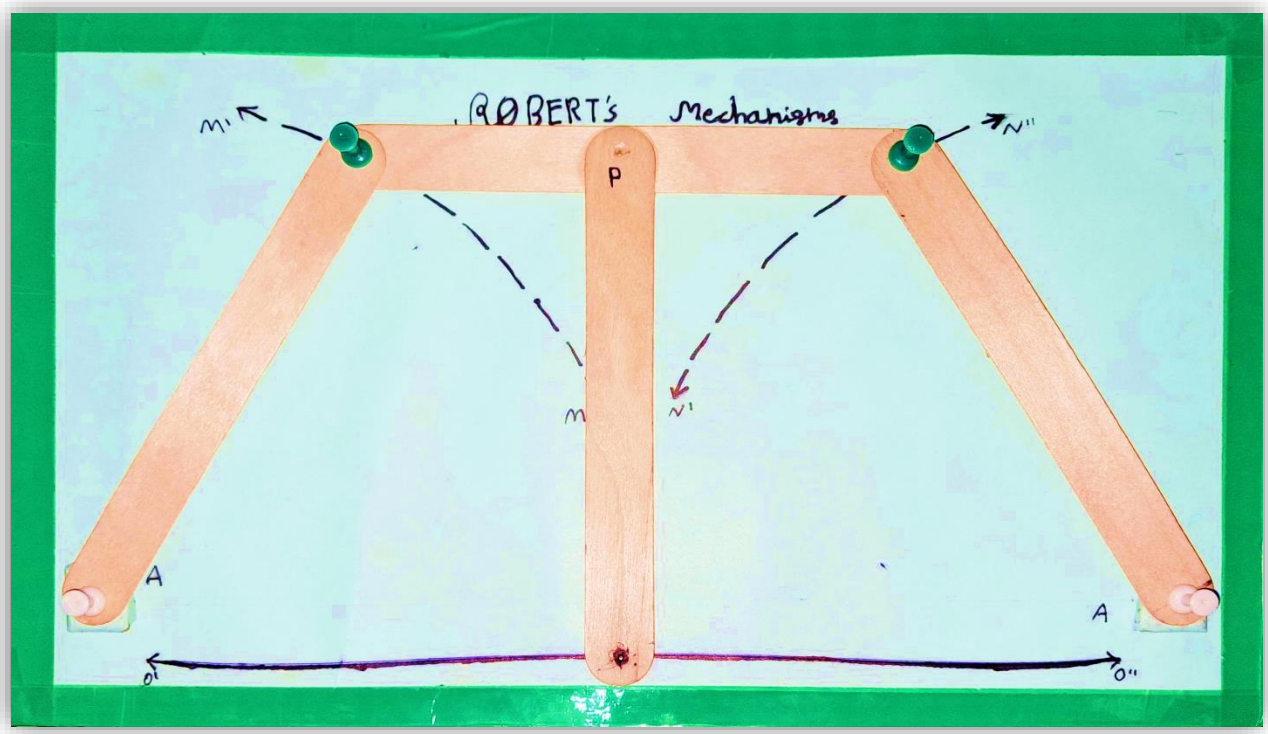


(a) Motion mode (b) Singularity position (c) Lockup mode

Figure 13: *Utilization of RLRBML: (a) Motion Mode, (b) Singularity Position, (c) Lockup Mode.*

The RLRBML provides a variety of potential applications including the linear motion guiding mechanism, linear actuator, linear positioning stage, drawing tools and pseudo rigid body model of the compliant translational joint in Micro-Electro-Mechanical Systems/Precision Engineering (Hao et. al., 2014).

OUR MODEL



Conclusion

Fabricating a Roberts Four-Bar Mechanism is a detailed process involving careful planning, precise calculations, and meticulous assembly. By following these steps, you can create a functional and reliable mechanism for various applications.

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